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Roll No

EC-5002-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Digital Communication

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Describe delta modulation. What are its limitations? How can they be overcome?
2. a) Explain Binary PSK signal with geometrical representation.
b) Explain MSK.
3. a) State and prove sampling theorem of Band Limited Signal.
b) Explain the generation of PWM and PPM signal.
4. a) Explain about matched filter and correlator.
b) Discuss Bayes theorem.
5. Consider a source with 8 messages having probabilities 0.22, 0.20, 0.18, 0.15, 0.10, 0.08, 0.05 and 0.02 respectively. Find the codes for messages using Huffman algorithm. Also calculate the efficiency of the codes.

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6. a) What do you understand from cumulative distribution function?
b) Write a short note on Power Spectral Density of Digital Data.
7. Draw block diagram of PCM system and explain it in detail.

OR

24 telephone channels, each band limited to 3.4 kHz, are to be time division multiplexed by using PCM. Calculate the bandwidth of the PCM system for 128 quantization levels and an 8 kHz sampling frequency.

8. Write short notes on any two of the followings
 - i) Huffman coding
 - ii) Channel capacity
 - iii) Entropy
 - iv) Information Capacity Theorem

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